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GIS and genetic algorithm based integrated optimization for rail transit system planning

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ABSTRACT

Planning a rail transit system is a complex process involving the determination of station locations and a network of lines linking those stations. This requires the consideration of various local conditions, including travel demand patterns, land use patterns, land values, topography and soil conditions. This is in addition to the many interrelated requirements and constraints related to the rail transit system stakeholders necessitating the application of optimisation models in order to achieve reliable and cost-effective rail transit systems. This paper develops an integrated optimisation model to simultaneously optimise the locations of stations and the associated line network linking them, using a geographic information system (GIS) and a genetic algorithm (GA). In addition to measuring the feasibility of station sites prior to the optimisation process, the developed model considers various local conditions and multiple planning requirements that arise from passengers, operators and the community.

The proposed model is tested for robustness through a real-world case study of the City of Leicester in the UK. The results show that it can generate station locations and line network alignment that satisfy the various stakeholder requirements and constraints. A comparative analysis is also conducted to reveal the importance of measuring the feasibility of station sites prior to the optimisation process so as to direct the search process towards optimal solutions, as well as to compare individual and simultaneous optimisation methods for planning a rail transit system. The developed model can be used to speed up the planning process and improve decision-making in respect to selecting a reliable and cost effective solution.

1. Introduction

The continuous growth of urban areas and associated need for mobility has stimulated the construction of new rail transit systems in many cities. Despite requiring substantial investment, such systems play a vital role in accommodating growth in travel demand, stimulating economic activities and providing fast, reliable and convenient service for those living on major corridors and seeking to access important activity areas. A basic problem with planning rail transit systems is determining the most economical locations of stations and configuration of rail lines linking the stations that can provide good service quality for passengers and benefits for the community. This requires the consideration of various local factors, including travel demand patterns, demography, topography and

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existing transport networks while satisfying a set of constraints that arise from passenger, operator and community requirements. The complex correlations and interactions between this multiplicity of requirements means that optimisation models are necessary to achieve reliable and cost-effective planning solutions.

In general, four main objective functions are used to solve the rail transit system planning problem: passenger costs, operational costs, construction costs and the population or traffic coverage. Passenger costs cover journey travel time from the origin to the desired destination, and associated financial costs which include the cost of using the rail transit service and the cost of access to/from rail transit stations (Chien and Schonfeld, 1997; Van Nes and Bovy, 2000). The operational costs of a rail transit system are measured in unit cost per passenger-kilometre (De Rus, 2012; ORR, 2012; TfL, 2015) which refer to how much it costs a train operating company, on average, to move a passenger over 1 km. Typically, the construction of rail transit systems includes the costs associated with ensuring the right-of-way of the lines and stations, performing earthworks, fastening rails, building structures (e.g. bridges, tunnels, stations) and other such items. Finally, some rail transit system planning studies incorporate maximum population coverage into the optimisation framework (Dufourd et al., 1996; Bruno et al., 2002), while some others embed maximum trip coverage (Laporte et al., 2005; Gutiérrez-Jarpa et al., 2013).

A number of studies have cast the rail transit planning problem within the context of multi-objective optimisation (e.g. Laporte et al., 2011; Gutiérrez-Jarpa et al., 2013; Rashidi et al., 2016), while others have formulated it within the context of a single-objective optimisation (e.g. Dufourd et al., 1996; Laporte et al., 2005; Lai and Schonfeld, 2010; Costa et al., 2016). Multiple optimisation entails planning the system so as to provide primarily either: (i) minimum passenger costs and maximum population or traffic coverage or (ii) maximum population or traffic coverage and minimum construction costs. Single-objective optimisation, meanwhile, entails planning the system to provide either: (i) minimum passenger or operator costs subject to a budget constraint; (ii) maximum population or traffic coverage subject to a budget constraint; or (iii) maximum population or traffic coverage or minimum construction costs without subjecting this to any constraints.

The focus of single-objective optimisation methods on a particular aspect has the limitation of not achieving optimality in rail transit system design in which there are multiple factors to consider simultaneously. This paper develops and tests an integrated multiple-objective optimisation model for both the locations of stations and the line network linking them, using a geographic information system (GIS) and a genetic algorithm (GA). The rest of the paper is structured as follows. Section 2 reviews the state-of-the-art in the development of models for rail transit system planning and highlights the associated gaps/limitations. Section 3 presents the framework of the proposed optimisation model. Section 4 demonstrates the application of the model in a real world case study. Section 5 presents a sensitivity analysis to demonstrate the importance of the initial identification of the potential station locations for directing the search process towards optimal solutions, and compares individual and simultaneous optimisation methods for planning a rail transit system. Section 6 evaluates the goodness of the solutions obtained by the developed optimisation model, with concluding remarks in section 7.

2. Literature review

Early studies on rail transit system planning mainly concentrated on the spacing between stations along a predetermined rail transit line. Vuchic and Newell (1968) developed an analytical model to determine spacing between stations in the case of a linear transit line where the served population commutes to one central point. With the objective of minimising passenger travel time, the model calculates interstation spacing by solving a set of simultaneous differential equations specifying the optimality condition. Since this model does not consider competitive transport modes, in a subsequent study, Vuchic (1969) proposed a similar model assuming a uniform distribution of passengers along the railway line, but taking account of competitive transport modes. The model calculates the optimal spacing between stations with the objective of maximising the number of railway passengers assuming that the passengers selected the system on the basis of shorter travel time. Subsequent work by Kikuchi and Vuchic (1982) developed a theoretical model to determine optimal station spacing and vehicle stopping policies for a rail transit system under different operating conditions. The objectives were minimum travel time and minimum total cost (user cost and vehicle operating cost) while considering factors like passenger volume, vehicle capacity, headway and access speed. The assumptions in these models of uniform distribution of the population along rail transit lines and commuting to a single central point are simplistic, however, and therefore these models have only limited practical applications.

Recent model improvements have involved more realistic assumptions and attempts to incorporate various planning and real world constraints. Dufourd et al. (1996) proposed a Tabu search heuristic to locate a rapid transit line with a known terminus linking a fixed number of stations on a grid network. With the objective of maximum population coverage, the model determines station locations, while the line alignment is generated according to the shortest path between the stations. Laporte et al. (2002) developed a model to locate a fixed number of stations on a predefined rail transit alignment with the objective to maximise population coverage, subject to a station spacing constraint. The model estimates the demand for each potential station by triangulation of census tracts, assuming that the percentage of the captured travellers decreases as their access distance increases. Schöbel et al. (2003) presented a model to locate additional stations along existing railway line networks. The objective of the model is to minimise the number (cost) of additional stations while ensuring coverage of all demand centres, assuming the traffic load for each railway line is known. The demand centre, which may represent activities such as a settlement area, shopping centre or a school, is assumed to be covered if the next station is within a specified radius from it. Schöbel (2005) extended the model to a bi-objective one where the maximisation of demand coverage is considered in parallel with the minimisation of the number of stations.

Schöbel and Scholl (2005) presented several models, using integer programming, to select a subset of lines from a line pool to connect several stations to minimise passenger inconvenience under a budget constraint. Passenger inconvenience is measured by the

travel time and number of transfers along the lines, where the cost of each line is assumed to be known a priori. Hamacher et al. (2009) also extended the Schöbel et al. (2003) model by proposing two objective functions. In the first objective, the number of stations is minimised such that all demand centres would be within a specified radius. In the second objective, the sum of the distances between demand centres and their closest stations is minimised while the number of new stations is fixed.

Laporte et al. (2011) developed a bi-objective model for determining optimal rail line networks and associated station locations. The model minimises a linear combination of the two objective functions, which consists of the construction cost and population covered by the network. Similarly, Gutiérrez-Jarpa et al. (2013) modelled and solved the problem of locating a rapid transit network in such a way as to maximise origin-destination traffic capture and minimise construction cost. A mixed integer formulation is presented to solve the problem, assuming a predefined topology of the network. To deal with both objectives, the problem is first solved for each objective separately, then a linear combination of the two objectives is minimised, corresponding to the straight line passing through the extreme solutions found. The construction cost of the network lines, meanwhile, is calculated as a linear function of their corresponding length and the captured traffic demands are computed in the absence of other competing transport modes. Rashidi et al. (2016) presented a mixed integer optimisation model to determine optimal station locations in a network. Assuming a predefined topology of the network, the model seeks to maximise the traffic covered by the stations and minimise the total budget allocated. The travel demand is estimated using origin destination matrices considering access time and in-vehicle travel time. Similarly, van der Hurk et al. (2016) presented an optimisation model in which they use mixed integer programming to select shuttle lines and frequencies for planned closures in high frequency urban transportation networks. The model is formulated to select shuttle lines and frequencies from a predetermined pool of candidate shuttle lines and frequencies that minimise passenger inconvenience cost, including transfers and frequency-dependent waiting time costs, within a budget for operating cost. Carrizosa et al. (2016) extended the Hamacher et al. (2009) model by seeking a set of additional stations covering all demand centres. Instead of minimising the number of additional stations, the additional passenger travel time is minimised.

Some studies on optimal locations of rail transit stations have integrated geographic information system (GIS) and artificial intelligence-based optimisation techniques such as genetic algorithms (GA), simulated annealing and ant-colony systems. For example, Jha and Oluokun (2004) developed a preliminary model based on GA and GIS to determine station locations along a given rail line alignment. The objective of the model is to minimise total system cost assuming that all travellers commute to one central point. Samanta and Jha (2011), presented a two-stage analytical model to optimise the station locations for a single rail transit line. Assuming that the line alignment constituted a linear connection between stations, the model applies GA to obtain optimal locations and sequences of stations. The objective of the optimisation is to minimise total system cost per person, which is a function of passenger travel time cost, system operation cost and construction cost (here, only the stations' right-of-way costs), while ensuring that station spacing remained constrained. Costa et al. (2016) presented a heuristic optimisation model based on a simulated annealing algorithm to optimise a high speed rail (HSR) alignment and associated station locations that minimise an objective function considering construction costs. Similarly, using a heuristic solution algorithm based on GA integrated with a background GIS database, Lai and Schonfeld (2016) proposed a model for simultaneously optimising a rail line alignment and associated station locations connecting a pair of predefined terminal stations. The model is formulated to take into account both the maximisation of trip coverage and the minimisation of the system construction cost while satisfying the general geometry requirement of the line alignment and station spacing constraints. Li et al. (2017) proposed a model combining bidirectional distance transform (DT) and GA to optimise railway alignments in complex mountainous regions. The model consists of two stages. In the first stage the model generates various railway paths using bidirectional DT scanning, and then selects the path that can generate a corridor at the lowest cost. In the second stage, the model searches for an optimised alignment in the corridor using GA. In a subsequent study, Pu et al. (2019) built on this model to optimise a railway alignment and associated station locations concurrently. The objective of the optimisation is to minimise construction and operational costs while satisfying constraints on the coupling between railway alignment and station locations. As with most previous studies, however, this model is designed for a single rail line and, therefore, is not appropriate for the planning of rail systems with multiple lines. In addition, it disregards the other crucial aspects (beyond construction and operation costs) of rail transit system planning, such as passenger cost, traffic demand and competitive transport modes.

The aforementioned models address many aspects of rail transit system planning but have a number of constraints that limit their application. These include: (1) a lack of a feasibility phase to identify potential station locations. The models assume that candidate sites for rail transit stations can be located anywhere in the study area or along the rail transit line without considering the various geometric, environmental, topological and budget constraints. A feasibility study would avoid potential problems including locating stations in areas that are restricted, infeasible or incur unnecessary costs; (2) inappropriateness in the context of a multi-line rail network as the methods are designed for a single rail line; and (3) failure to account for the complex interactions between rail lines and station locations by treating line alignment and station location in separate optimisation processes.

This paper addresses these limitations by developing an integrated optimisation model using GA integrated with GIS tools to optimise the locations of stations in tandem with the line network linking them. Furthermore, the developed model measures the feasibility of station sites to determine a candidate pool of feasible station locations, and considers various local factors and the multiple planning requirements that arise from passengers, operators and the community. This model aims to help rail transit planners to expedite the planning process and improve decision-making in respect to selecting reliable and cost-effective solutions. The main contributions of this paper are:

1. A new integrated optimisation model for planning a rail transit system that accounts for the various local conditions and requirements that arise from the different stakeholders (i.e., passengers, operator and the community). This provides most benefit to the area it serves through the mitigation of congestion and improvements in mobility, economic development and the environment.

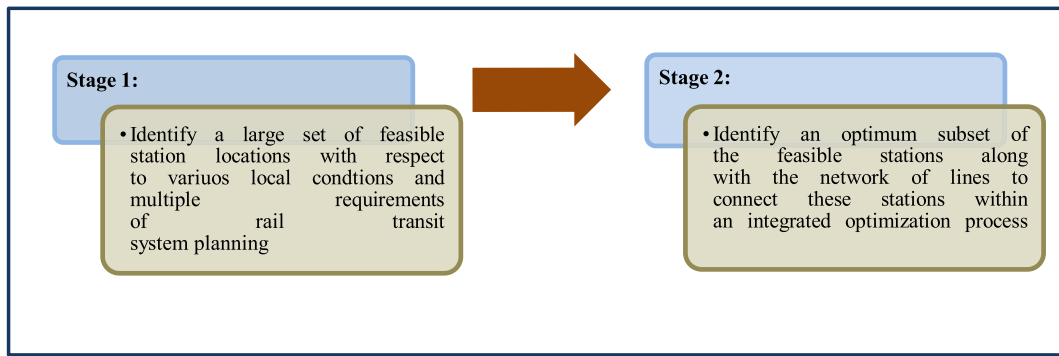


Fig. 1. Stages of the proposed integrated optimisation model.

2. Consideration of the complex correlations and interactions between rail transit station locations and line alignments by integrating these two tightly linked components in a single optimisation process, in addition to the simultaneous optimisation of multiple rail lines. This is important for enhancing the system performance and minimising the overall system costs, and thus achieving a cost-effective solution.
3. Introduction of a model formulation process that addresses the essential trade-off between the objective of maximising rail system usage and minimising passenger travel time on the one hand, and the objective of minimising the construction cost on the other hand, while complying with various constraints. This trade-off is essential for the acquisition of a rail transit system that can provide high service quality and benefits for both passengers and the whole community, with acceptable capital and operational costs.

3. The model framework

Since it is impractical to check the feasibility of every point in the study area as a potential station location, the model is broken down into two stages, as shown in Fig. 1. The first stage screens the study area to identify a pool of feasible station locations based on a comprehensive consideration of various local conditions and the multiple requirements of rail transit system planning. The second stage selects a subset of feasible stations from that pool and generates a network of lines to link these stations within the context of an integrated optimisation process to realise the final system configuration. The optimisation framework is designed to accommodate complex correlations and interactions between the multiple requirements of the rail transit system stakeholders (passengers, operators and the community) and to make trade-offs between them to achieve an optimal and cost-effective system. Although the model framework is specified for underground/metro rail systems with stations at ground level, it is adaptable to other variations of rail transit system planning.

3.1. Stage 1: determination of potential rail transit station locations

This stage applies the following four main steps to generate the feasible rail transit station locations within the context of a GIS-based algorithm:

Step 1: divide the study area into cells and create a GIS layer (Ω_s) with attributes:

C^i feasibility of cell i to be a station (0- infeasible, 1-feasible)

It is assumed that each of the generated cells represents a potential location for a rail station. The size of each cell, therefore, should represent the typical size of a rail station.

Step 2: evaluate the locations of each of the cells generated in Step 1 with respect to the various local conditions and requirements of rail transit system planning, using their predefined respective datasets in ArcGIS. This consists of determining the following in respect to an area within a defined walking distance of each potential station site: (i) number of airport terminals and existing rail stations, (ii) number of car parking facilities, (iii) number of bus stops, (iv) average area of land devoted to commercial landuse, (v) average population density, (vi) average land value, and, (vii) average proportion of land that is deprived or undeveloped. The use of the average values in some of these parameters, instead of using standard deviations, is based on assuming a uniform distribution of their values within the defined walking distance to the corresponding potential stations.

Walking distance refers to access/egress distance to/from rail stations. It is relevant to planning and design because of its influence on the use of rail services, and social and economic activities in the proximity of stations. Therefore, it provides a standard basis for a comparative evaluation of the potential station locations. However, the specification of the actual walking distance can be variable ranging from 400 to 800 m depending on factors such as operational environment (e.g. urban or sub-urban), and service quality at the rail station (e.g. network connectivity, area coverage and service facility and frequency) (O'Sullivan and Morrall, 1996; Evans et al., 2007; Bertolini, 1999; Reusser et al., 2008; Zemp et al., 2011; Kamruzzaman et al., 2014). Because of the variability, the model framework in this paper treats walking distance as a parameter to be specified by planners.

Step 3: find all station cells that intersect with environmentally sensitive areas (e.g. historic buildings, national parks, woodlands,

forests, rivers and sites of scientific interest) and exclude them from the search space of potential station locations. This is done by laying the generated station cell layer (Ω_C) over the environmentally sensitive areas layers (Ω_e) to generate the feasible grid layer for stations (Ω_f).

$$\Omega_f = \Omega_C \cap \overline{\Omega_e} \quad (1)$$

Step 4: identify all the cells (C_i) within the feasible station layer (Ω_f) that satisfy predefined threshold values for each of the requirement evaluation criteria defined in Step 2 and assign them with integer values, which represent the weight of the satisfied criteria, as follows:

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum AT \middle| D_w \right) \geq N_{AT} >, 0 \text{ otherwise} \quad (2)$$

Where: AT is the number of airport terminals within the specified walking distance, D_w is the walking distance and N_{AT} is the pre-specified threshold number of airport terminals within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum RS \middle| D_w \right) \leq N_{RS} >, 0 \text{ otherwise} \quad (3)$$

Where: RS is the number of existing railway stations within D_w , and N_{RS} is the pre-specified threshold number of existing railway stations within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum CP \middle| D_w \right) \geq N_{CP} >, 0 \text{ otherwise} \quad (4)$$

Where: CP is the number of car parks within D_w , and N_{CP} is the pre-specified threshold number of car parks within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum BS \middle| D_w \right) \geq N_{BS} >, 0 \text{ otherwise} \quad (5)$$

Where: BS is the number of bus stops within D_w , and N_{BS} is the pre-specified threshold number of bus stops within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\overline{PD} \middle| D_w \right) \geq V_{PD} >, 0 \text{ otherwise} \quad (6)$$

Where: $\overline{PD}$ is the average population density within D_w , and V_{PD} is the pre-specified threshold value of average population density within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum \overline{CA} \middle| D_w \right) \geq V_{CA} >, 0 \text{ otherwise} \quad (7)$$

Where: $\overline{CA}$ is the average commercial landuse area within D_w , and V_{CA} is the pre-specified threshold value of average commercial landuse areas within D_w . Commercial landuses include office complexes, retail areas and recreational centres.

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\sum \overline{DA} \middle| D_w \right) \geq V_{DA} >, 0 \text{ otherwise} \quad (8)$$

Where: $\overline{DA}$ is the average area of deprived or undeveloped land within D_w needing regeneration (i.e. land that has a strong potential for future economic development). V_{DA} is the pre-specified threshold value for the average area of deprived/undeveloped land parcels within D_w .

$$\text{For each } C_i \in \Omega_f : C^i = 1, \text{ if } < \left(\overline{LV} \middle| D_w \right) \leq V_{LV} >, 0 \text{ otherwise} \quad (9)$$

Where: $\overline{LV}$ is the average land value within D_w , and V_{LV} is the pre-specified threshold value of the maximum allowed land acquisition value (unit cost per unit area) for stations.

The threshold values are defined by rail transit planners depending on the demographic, topological, economic and land use pattern characteristics of the potential service area. The values are critical because they not only affect the number and locations of the candidate pool of potential stations, but also the entire configuration of the proposed rail system and subsequently the achievement of the desired system objectives. In general, the higher the threshold values, the better the satisfaction of the system requirements. On the other hand, setting high threshold values may result in a significant reduction of the search space for the optimisation model with the consequence of a sub-optimal solution. Section 5.1 investigates the effect of different threshold values on the optimisation search space in order to determine threshold values that would not result in the exclusion of potential optimal station locations.

3.2. Stage 2: Optimisation of station locations and associated line network

Once the set of feasible station locations is determined using the GIS-based algorithm, the second stage employs an integrated optimisation process to generate the final configuration. It simultaneously selects the best subset from the potential station set identified in the first stage along with the best line network to link the selected stations. Fig. 2 illustrates the framework of the proposed

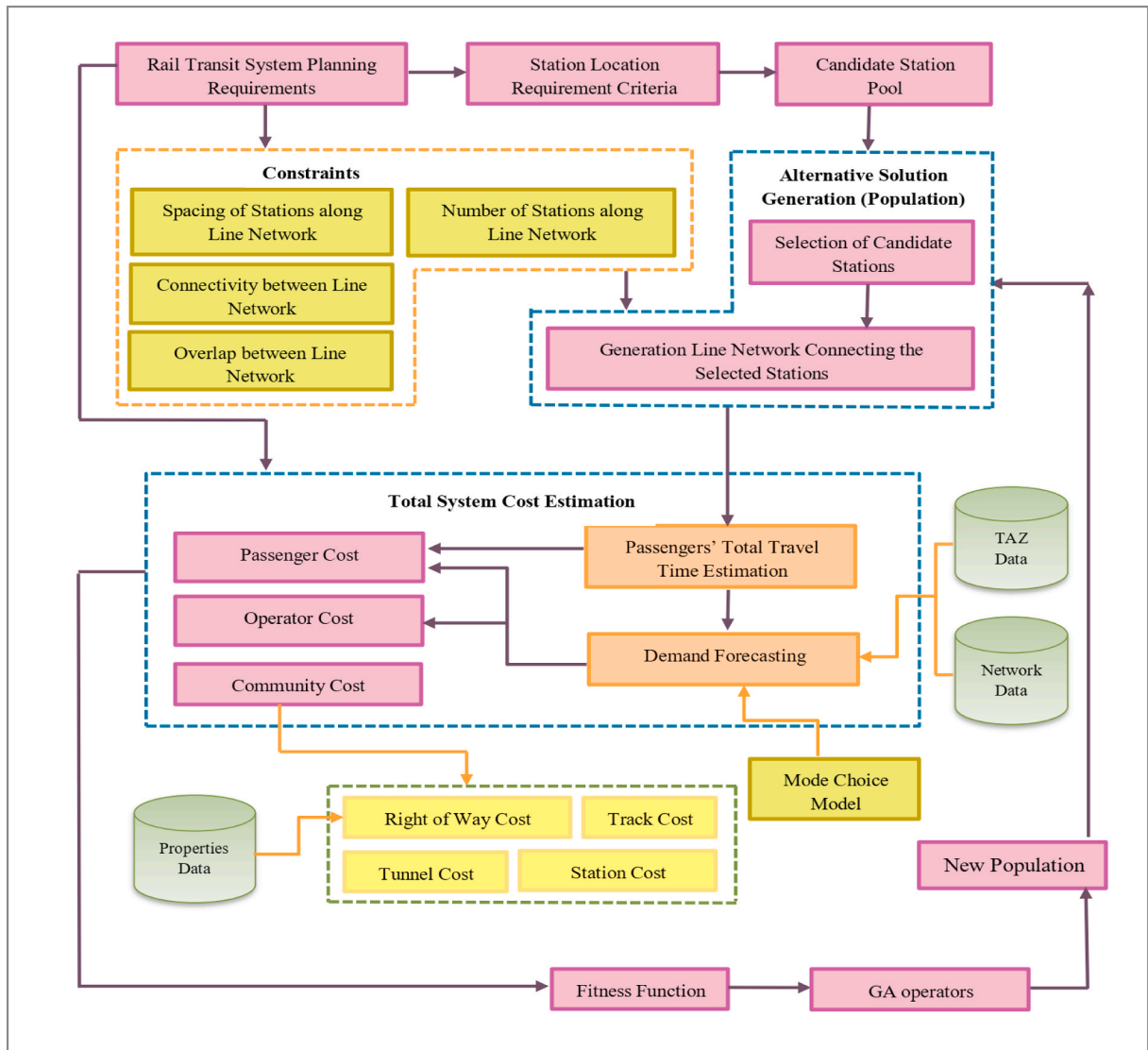


Fig. 2. Framework of the integrated optimisation algorithm.

integrated optimisation algorithm which is designed to simultaneously determine the optimal locations of stations and the best line network. As depicted in Fig. 2, a heuristic solution algorithm based on GA is employed to seek a solution that best fits the desired objectives of the three interconnected stakeholder requirements (passenger, stakeholder and community costs). A GA is an evolutionary optimisation technique widely utilised for solving various complex and large-scale problems. Following the concept of biological evolution, GAs evolve a population of different alternative solutions towards the optimal solution. The evolution begins with a population of random alternative solutions and then through the course of a number of generations/iterations the fitness of the population is evaluated with respect to the desired objectives of the problem. Thereafter, a number of individuals are selected stochastically from the current population based on their fitness values and modified with genetic operators (crossover and mutation) to produce a new population for the next generation/iteration. This process is continued until either a specified number of generations, which is fixed a priori, has elapsed, or other predetermined convergence criteria are met (Goldberg, 1989). The following subsections detail the key steps of the proposed GA-based optimisation algorithm.

3.2.1. Generation of the alternative solutions

The generation of the alternative solutions consists of selecting a set of stations from the candidate station pool identified in section 3.1 and generating a network of lines connecting the selected stations while satisfying predefined optimisation constraints: (i) minimum and maximum number of stations along each line of the network ($N_{s_{min}}$ and $N_{s_{max}}$); (ii) minimum and maximum distance between any two successive stations along each line of the network (Δs_{min} and Δs_{max}); (iii) minimum number of intersected/transfer stations along each line in the network ($C_{n_{min}}$); and, (iv) maximum overlapped or common sections between any two lines in the

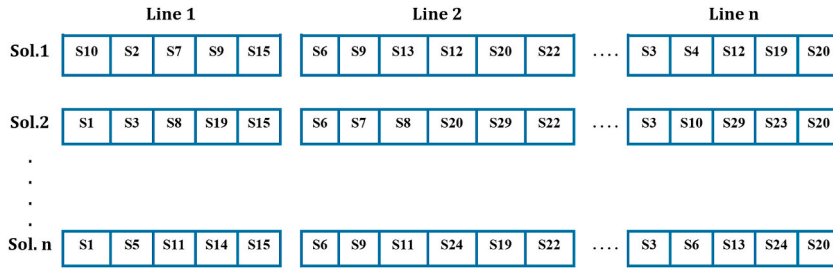


Fig. 3. Generated alternative solutions.

network (On_{max}).

It is assumed that, depending on the major traffic flow patterns of the area/city to be served, rail transit planners are able to determine the number of lines desired and the terminal stations of each line (i.e., the beginning and end points of each line). It is notable that the generated alternative solutions largely depend on the set of terminal station pairs and constraints applied for checking the feasibility of the generated solutions. The planner's knowledge is therefore vital to the acquisition of an optimal rail transit system. In GAs, a set of candidate solutions is called a population; each candidate solution in the population pool is known as a chromosome or an individual; each decision variable of the solution is known as a gene; and each parameter of the decision variable is known as a bit. Accordingly, the proposed heuristic algorithm treats the alternative solutions as chromosomes, associated rail lines as genes, and associated stations of the lines as bits. Fig. 3 illustrates the schematic diagram for the generated alternative solutions.

3.2.2. Fitness function

The proposed optimisation algorithm evaluates the fitness of each alternative solution in terms of minimisation of cost. Based on the rail transit system planning requirements, the fitness value of each alternative is evaluated with respect to three main cost components; passenger, operator and community.

$$T_c = \phi_p P_c + \phi_o O_c + \phi_c C_c$$

Where: T_c is the total system cost, P_c , O_c , C_c are the passenger, operator and community costs respectively over the life span of the proposed rail system, and ϕ_p , ϕ_o and ϕ_c are the respective coefficients of passenger, operator and community costs. These coefficients provide flexibilities into the proposed algorithm to allow rail transit planners to prioritise a particular stakeholder requirements over others when required. The remainder of this section details the mathematical formulations of each of these three cost components.

- (i) Calculation of passenger cost: this is the time cost difference between utilising a train, instead of the bus and car. This difference signifies how much the cost of passengers' travel time can be decreased or saved by using the proposed rail system instead of using car or bus transport modes. The annual passenger cost (A_p), assuming daily services with demand variations over weekdays and weekends, is:

$$A_p = (\phi_d \times 365) \times \sum_{t=1}^q \left(\sum_i \sum_j (P_{ij} \times T_{Ctij} - P_{cij} \times T_{Ctcij}) + (P_{ij} \times T_{Ctij} - P_{bij} \times T_{Ctbij}) \right) \quad (10)$$

Where:

ϕ_d is the coefficient to account for demand pattern variations over weekdays and weekends;

q is the number of lines in the network;

P_{ij} is the number of railway passengers from zone i to zone j in time period t ;

T_{Ctij} is the cost (£) of the total travel time by train from zone i to zone j in time period t ;

P_{cij} is the number of railway passengers from zone i to zone j in time period t that may switch to car if the rail system is not available;

T_{Ctcij} is the cost (£) of the total travel time by car from zone i to zone j in time period t ;

P_{bij} is the number of railway passengers from zone i to zone j in time period t that may switch to bus if the rail system is not available;

T_{Ctbij} is the cost (£) of total travel time by bus from zone i to zone j in time period t .

The demand values are forecasted using a discrete mode choice model, which is incorporated in the proposed optimisation algorithm framework. The origin-destination trip matrix at zonal level for each alternative transport mode (car, bus, and active mode) is taken as known from an external regional demand forecasting model. It is assumed that the trip distribution pattern between the traffic analysis zones of the study area does not change after building the proposed rail system. Therefore, for each particular zone pair, the total demand matrix is computed by summing up the corresponding demand matrices of the three existing transport modes. The proposed optimisation algorithm uses this demand matrix as an input and splits it into four matrices, one for each mode (rail, car, bus and active mode). This is via the incorporated logit choice model, the most widely used discrete choice model in real world practice. In addition to the consideration of the characteristics of travellers, the logit choice model incorporates sets of variables that are related to the attributes of alternatives. These variables involve in-vehicle travel time, access time, egress

time, waiting time, search time for parking spaces, travel fares, fuel and other operating and maintenance costs.

The total train time (T_{Ctt}) includes passengers' access time to/from station, waiting time at station and on-train travel time. Thus, (T_{Ctt}) is calculated as follows:

$$T_{Ctt} = T_{ta} \times A_0 + T_{tw} \times W_0 + T_{tt} \times T_0 \quad (11)$$

Where:

T_{ta} is the passenger's access time to/from the station (min);

A_0 is the unit costs of access time (£/min);

T_{tw} is waiting time at station (min);

W_0 is the unit costs of waiting time (£/min);

T_{tt} is on-train travel time (min) and;

T_0 is the unit cost of on train/in-car travel/on-bus time (£/min).

T_{ta} is a function of passenger walking distance to/from stations and speed. The distance (D_a) is computed by measuring the artificial links between the centroid of the traffic analysis zones (TAZs) of the study area and potential station locations. T_{ta} is then calculated using equation (12) where V_a is the passenger walking speed (V_a).

$$T_{ta} = \frac{D_a}{V_a} \quad (12)$$

The waiting time (T_{tw}) depends on the scheduled frequency of trains and their reliability, and passengers' arrival pattern at the boarding station. Since none of these elements can be evaluated accurately at this stage of the planning process, T_{tw} is assumed to be equal to half of the train headway (H_w), which is a function of train frequency. It should be noted, however, that this assumption only holds for headways shorter than 10 min (Luethi et al., 2006; Frumin and Zhao, 2012; Zhao et al., 2013; Ingvardson et al., 2018).

$$T_{tw} = \frac{1}{2} \times H_w \quad (13)$$

The on-train travel time (T_{tt}) depends on the distance between boarding and alighting stations (D_s), average train speed between boarding and alighting stations (V_t), and dwell time at calling stations (n_s), which can be represented as a fixed value (T_d). Thus, T_{tt} is calculated as:

$$T_{tt} = \frac{D_s}{V_t} + (n_s \times T_d) \quad (14)$$

The travel distance between each boarding and alighting station in the network (D_s) is computed by implementing the Dijkstra's shortest path algorithm, which is embedded into the evaluation framework of the proposed optimisation algorithm. In this paper, the weight assigned to the lines linking the selected candidate stations is defined as their corresponding length while the algorithm searches for the shortest path between each boarding and alighting station pair, and thus the minimum travel distance between each boarding and alighting station (D_s) is obtained.

The total car time (T_{Ctc}) consists only of in-car travel time between origin and destination and the time spent searching for a parking space, and thus (T_{Ctc}) is calculated as:

$$T_{Ctc} = T_{ct} \times T_0 + T_{cs} \times S_0 \quad (15)$$

Where:

T_{ct} is the in-car travel time (min);

T_{cs} is the search time for a park space (min) and;

S_0 is the unit cost of search time for a parking space (£/min).

The total bus time (T_{Ctb}) consists of passenger access time to/from bus stop, waiting time at the bus stop and on-bus travel time.

$$T_{Ctb} = T_{ba} \times A_0 + T_{bw} \times W_0 + T_{bt} \times T_0 \quad (16)$$

Where:

T_{ba} is the passenger's access time to/from bus stop (min);

T_{bw} is the waiting time at bus stop (min) and;

T_{bt} is the on-bus travel time (min).

The road and bus network travel time information between each traffic analysis zone pair are assumed to be available from relevant transport agencies for the AM/PM peak, intermediate peak (IP) and off peak (OP) periods, thus taking into account traffic congestion levels for different time periods.

The rail transit network usage and its distribution pattern over the existing transport modes (i.e. car and bus) in cases where rail is not available are predicted with the logit mode choice model.

Since the passenger cost recurs throughout the lifespan of the system, a life cycle analysis is performed to calculate the passenger cost of the system over its design life. Thus, assuming a fixed annual interest rate, i , and a design life of n years, the passenger cost over the system life cycle (P_C) is:

$$P_C = A_P \frac{(1+i)^{n-1}}{i(1+i)^n} \quad (17)$$

- (ii) Calculation of the operator cost: this is defined as the difference in the operation and maintenance cost differences between utilising rail and other transport modes, specifically car and bus. This difference signifies how much the total cost of operations and maintenance can be decreased or saved due to the use of the proposed rail system instead of using either car or bus transport modes. The operation and maintenance costs of each mode are a function of the corresponding travelled distance and potential passengers. Hence, the annual operating cost (A_O) is calculated as follows, assuming daily services with demand variations over weekdays and weekends:

$$A_O = (\phi_d \times 365) \times \sum_{l=1}^q \left(\sum_t \sum_i \sum_j ((P_{ij} \times L_{tij} \times M_{t0} - P_{cij} \times L_{cij} \times M_{c0}) + (P_{ij} \times L_{tij} \times M_{t0} - P_{bij} \times L_{bij} \times M_{b0})) \right) \quad (18)$$

Where:

L_{tij} is the distance travelled by a train in km from station i to station j in time period t ;

M_{t0} is the unit operational and maintenance cost for the train (£/passenger-km);

L_{cij} is the car km travelled distance from zone i to zone j in time period t ;

M_{c0} is the unit operational and maintenance cost for car (£/passenger-km);

L_{bij} is the distance travelled by a bus in km from zone i to zone j in time period t and;

M_{b0} is the unit operational and maintenance cost for bus (£/passenger-km).

The train travelled distance between each stations pair in the network is calculated using the Dijkstra's shortest path algorithm. The bus and car travel distance between each traffic analysis zone pair are assumed to be available from the relevant transport agencies at different time periods. The demand values are forecasted using the logit mode choice model.

Similar to passenger cost, the operator cost can be calculated for the system life span (O_C) as:

$$O_C = A_O \frac{(1+i)^{n-1}}{i(1+i)^n} \quad (19)$$

- (iii) Calculation of the community cost: this is defined as the required capital cost for constructing stations and the line network connecting the stations, which is classified into location-dependent (C_{LDep}) and size-dependent (C_{SDep}) categories. The former involves those construction items which change significantly with different locations for the same station size, total line network length and construction type. The second category includes those construction items that do not change with different locations for the same station size, total line network length and construction type. Therefore, the costs for building and equipping stations with the required facilities and those for laying track and fastening rails can be listed under the size-dependent construction cost category, while land acquisition and tunnel costs belong to location-dependent construction costs. Thus, the community cost (C_c) is:

$$C_c = C_{LDep} + C_{SDep} \quad (20)$$

$$C_{LDep} = \sum_{l=1}^q (Rw_l + Tu_l) \quad (21)$$

$$C_{SDep} = \sum_{l=1}^q (Sb_l + Tr_l) \quad (22)$$

Where:

Rw_l is the right of way (i.e., land acquisition cost) for stations along line l (£),

Tu_l is the cost of building the tunnel structure along line l (£),

Sb_l is the cost of building and equipping the stations with necessary facilities along line l (£),

Tr_l is the total cost of laying rail tracks and fastening rails along line l (£).

Although these construction cost parameters are set for underground rail transit systems, they can be modified for over-ground systems by including the relevant construction parameters, such as the cost of building bridges. C_c represents the community cost over the system lifespan as the construction cost is only incurred at the onset of the project.

The right of way cost includes land acquisition and property damage cost items for building stations and the associated line

network. It is a function of where the stations are built and the associated line network constructed. Since this paper focuses on rapid rail transit system/metro, the cost of acquiring right of way depends only on the locations of the stations, because the deep tunnels through which the lines connecting the stations are passed entail no right of way cost. This construction cost item is computed via interaction with the GIS, in which the unit cost database of land parcels and properties are stored. The proposed optimisation algorithm first identifies all sections of land parcels and properties that are located within the stations' right of way boundaries, and then extracts the corresponding costs of these sections from the GIS database to compute the right of way acquisition cost. The right of way acquisition cost is the sum of the cost values of these land parcels and property sections.

The cost of building tunnel structures (T_u) is assumed to be a function of tunnel length and is calculated as:

$$T_u = \sum_{l=1}^q (T_L \times C_{T_u}) \quad (23)$$

Where:

T_L is the tunnel length of line l (km), and

C_{T_u} is the unit cost of building tunnel structure (£/km).

The cost of building and equipping a station (S_b) is a function of the size and number of stations along the line network:

$$S_b = \sum_{l=1}^q (N_s \times C_{S_b}) \quad (24)$$

Where:

N_s is the number of stations along line l , and

C_{S_b} is the unit cost of building and equipping a station with required facilities (£).

The cost of constructing the rail track (T_r) includes that of laying the tracks and fastening rails. It is a linear function of the line network length:

$$T_r = \sum_{l=1}^q (Len_l \times C_{T_r}) \quad (25)$$

Where:

Len_l is the length of line l (m), and,

C_{T_r} is the unit cost of laying rail tracks and fastening rails (£/m).

It is important to note that through the passenger cost function, the model tries to deliver adequate service quality for railway passengers by either decreasing passengers' rail travel time compared to bus and car travel times, or by increasing railway usage, or a combination of the two. Through the operator cost function, the model tries to increase the railway system revenue and achieve efficient system operation. Through the community cost function, the model tries to minimise the construction costs. The proposed integrated optimisation model therefore resolves the trade-offs by increasing the number of railway users in order to reduce both the proportional passenger and operational costs, while not significantly increasing the construction costs.

3.2.3. Evolution of the alternative solutions

As illustrated in Fig. 2, once a population of alternative solutions is initialised and the corresponding fitness is evaluated, the GA starts the evolution process to direct the population towards the optimum solution. Typically, GAs implement the evolution by applying three genetic operators; selection, crossover and mutation. In general, these operators, in particular the crossover and mutation operators, are problem specific, and must therefore be designed to fit the problem under investigation in order to facilitate the efficiency of the convergence towards a local optimum solution. Consequently, this paper proposes three genetic operators that fit the framework of rail transit system planning.

- (i) Selection operator: it determines which individuals are to be selected to form a mating pool for the next generation. The primary objective of this operator is to improve the average fitness of a population by giving individuals of higher fitness a higher chance to be selected for the next generation. Consequently, it directs the search to explore and exploit promising areas in the search space. The most widely used selection methods are roulette wheel selection, tournament selection and rank selection. Although each of these methods calculates the selection probability of individuals differently, they are all based on the same principle, which is survival of the fittest (i.e. fitter individuals have a higher chance of being selected for the next generation than weaker ones). Obviously, since different selection methods influence the performance of GA differently, close attention should be paid to choosing a selection method that ensures that the algorithm performs well, and this is usually evaluated in terms of

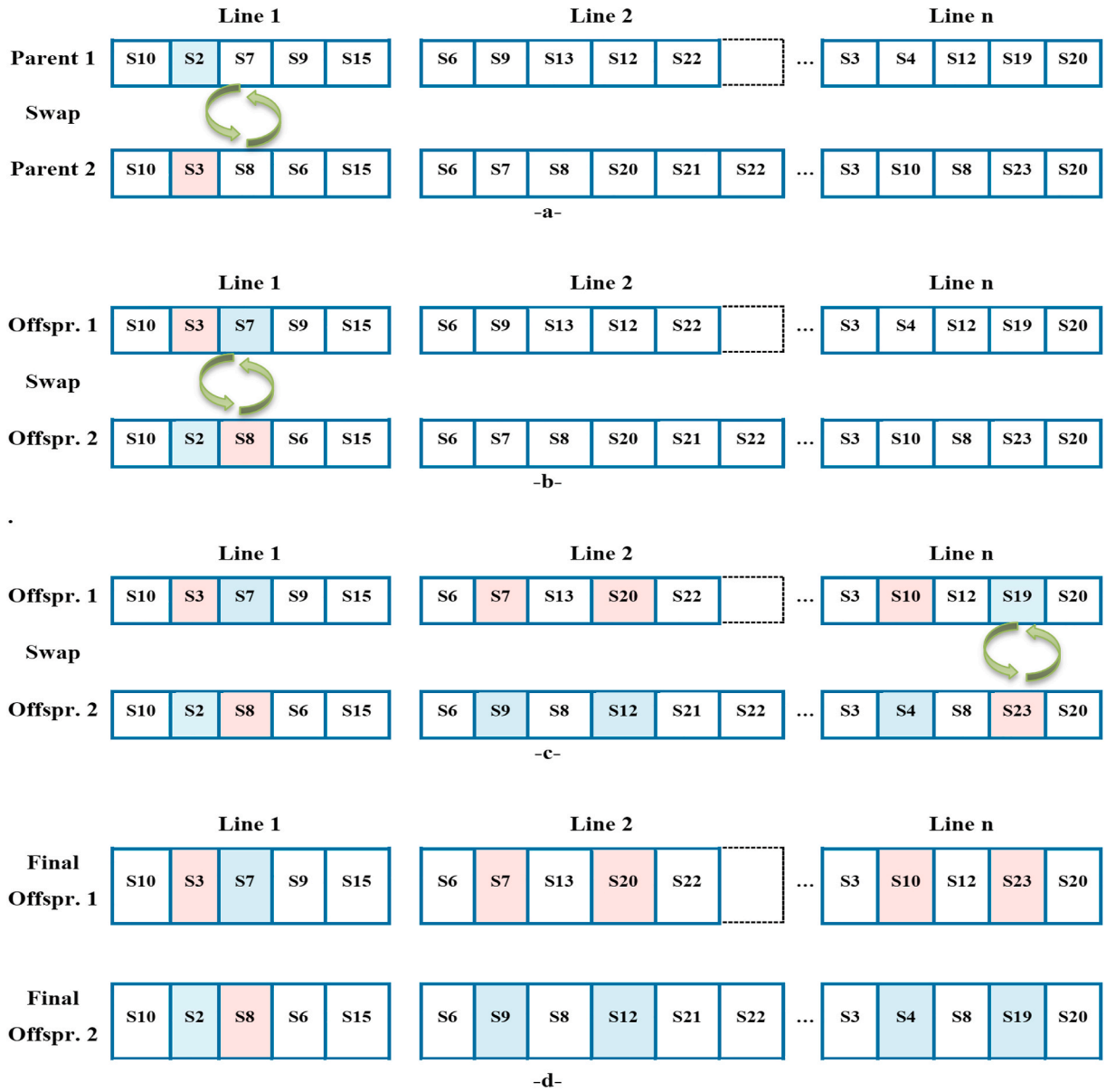


Fig. 4. Illustration of the uniform crossover approach.

convergence rate and the required number of generations to arrive at the optimal solution. In this paper, however, since the fitness values of the solutions can be very close, the tournament selection method is applied as it is sensitive enough to differentiate solutions of very similar fitness values. Tournament selection picks a subset of two solutions from the population randomly and then selects the best solution from this subset.

- (ii) **Crossover Operator:** this operator merges the individuals picked up by the selection operator to produce new individuals. Crossover operates on two individuals (parents) at a time and generates two offspring/children by exchanging the genetic information of both individuals (parents). The crossover operation occurs at a probability rate denoted as P_c , which is defined as the probability of the number of generated offspring in each generation to the population size. This means that not all the selected individuals undergo the crossover operation unless the P_c value is equal to one. A very high P_c value tends to explore more of the search space and mitigates the risk of converging to a local optima. On the other hand, it may result in a lot of computation time being wasted exploring unnecessary areas of the search space. It is, therefore, very important to utilise a P_c value which maximises the probability of the algorithm finding good solutions while converging speedily. The optimal value of P_c is problem specific (Patil and Pawar, 2015), and, accordingly, in section 5.2, this paper examines different P_c values to determine the optimal value for solving the rail transit system planning problem.

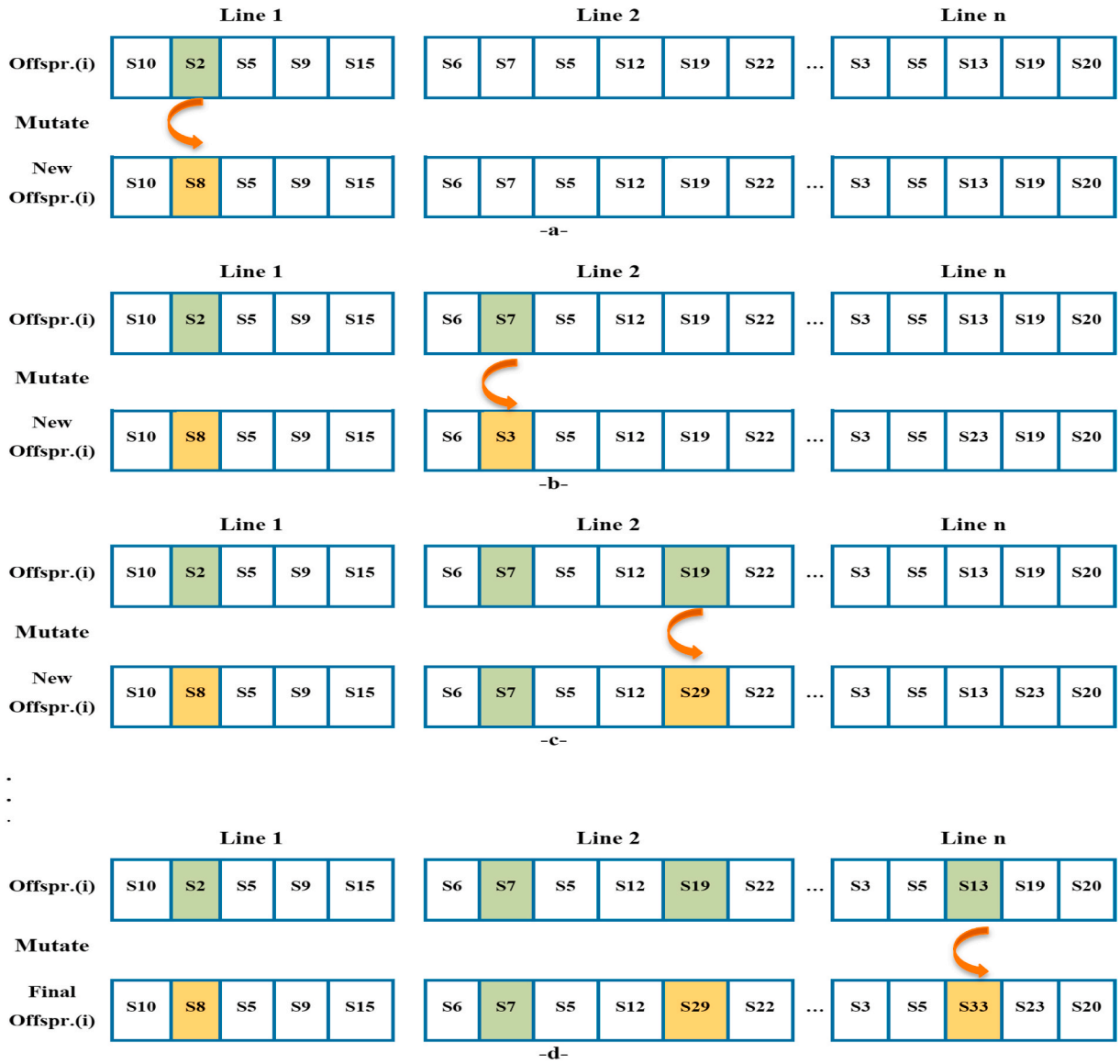


Fig. 5. Illustration of the mutation operator.

The most widely used crossover methods include one-point crossover, two-point-crossover and uniform-crossover. In this paper, adapting uniform crossover methods, the crossover operator is designed to merge parents at gene bit (station) level to produce new offspring. That is, it exchanges the corresponding stations of each two alternative solutions picked up for crossover individually, as illustrated in Fig. 4. The crossover operator is designed in such a way as to reject parental matings that result in infeasible offspring violating the defined constraints. This is by returning the mated gene bits back to their initial state before the mating and then mating the next gene bits, as illustrated in 4b and 4c.

- (iii) **Mutation Operator:** this operator alters the value of one or more genes of the offspring pool generated by the crossover operator and alters a part of their content. It is used to produce new genetic information to ensure that the algorithm diversifies over the search space so as to avoid the premature convergence towards suboptimal solutions. The mutation operator works on a single individual/offspring at a time and produces a new individual/offspring, and it occurs at a probability rate denoted as P_m . The mutation rate, or mutation probability, controls the frequency with which the mutation operator is applied. In general, a very high value of P_m tends to lead to a significant diversification of the search space. This may, in turn, pull good solutions away from the population and thus prevents the population from converging at any optimal solution. On the other hand, if the value of P_m is very low, the search mostly converges prematurely towards suboptimal solutions due to the higher exploitation rate. It is therefore very important to utilise a P_m value that maximises the probability that the GA finds the global or near-global

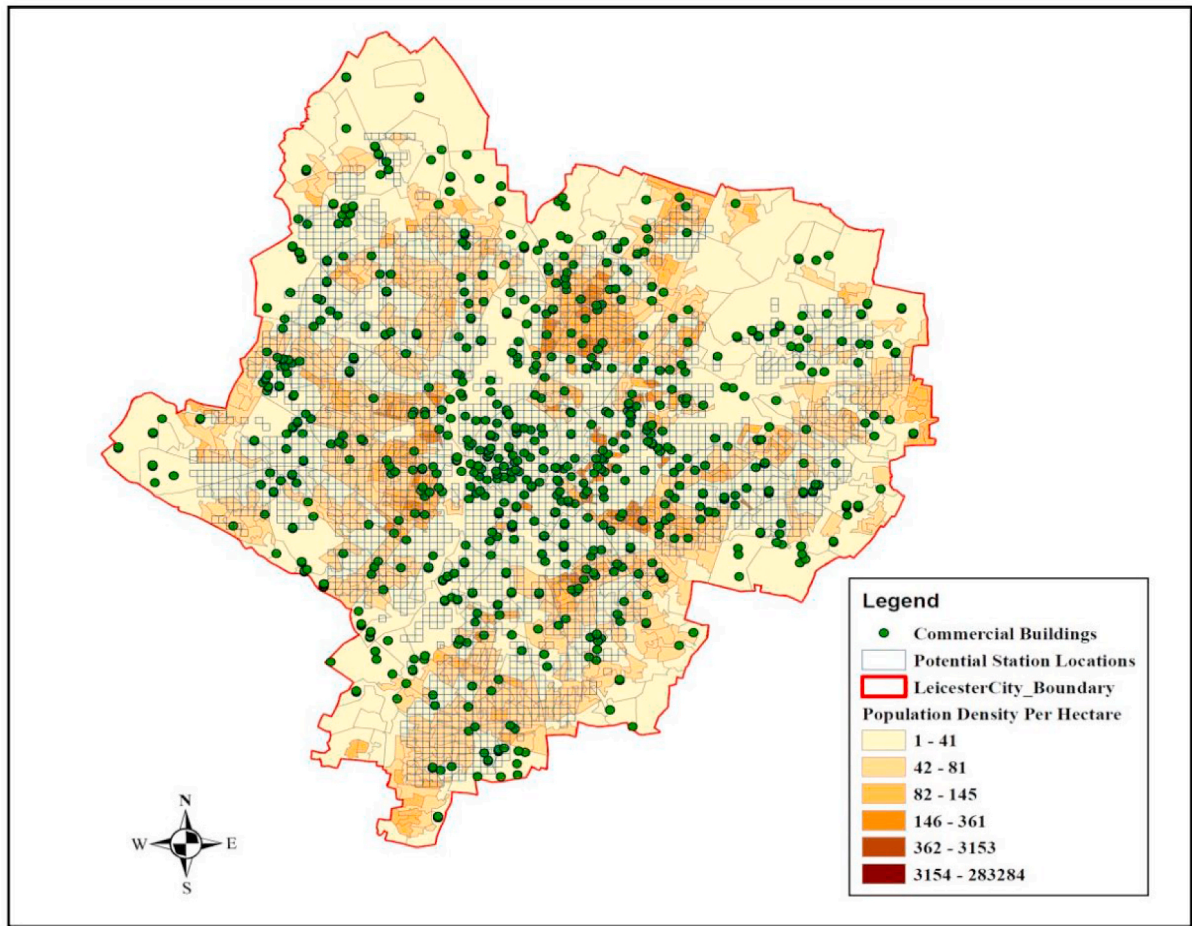


Fig. 6. Feasible station location solutions.

optimum solution of the problem. Since the optimal value of P_m varies for different problems, there is no consistent optimal value of P_m for most real world problems (Srinivas and Patnaik, 1994; Piszcz and Soule, 2006; Patil and Pawar, 2015). This paper therefore tests different values of P_m in section 5.3 to find the optimal P_m value to solve the rail transit system planning problem.

This operator is designed to take individuals from the offspring pool and randomly mutate them at the gene bit level (station) to produce new offspring (see Fig. 5). It individually selects a number of gene bits of the individual's chromosome with probability P_m , and replaces them with randomly selected gene bits to produce a new offspring. Since the mutation might produce infeasible offspring due to violation of the required constraints, the mutation operators are designed to reject the gene bit mutation that results in infeasible offspring. This is by returning the mutated gene bit back to its initial state before the mutation, and then mutating the next gene bit (see Fig. 5b and c).

3.2.4. Determination of the optimal solution

As depicted in Fig. 2, after the selection, the crossover and mutation operators (i.e., GA operators) are applied to the initial population so that a new population is formed for the next generation. In general, two stopping criteria are employed in GAs, objective based and generation based. In this paper, if one of the following two criteria is satisfied the evolution process is terminated and the best individual in the current population is identified as the optimal solution.

1. if the best individual of the population is not further improved over a specified number of successive generations, which is defined by the potential user/planner.
2. if the maximum number of generations is reached. This value is also defined by the potential user/planner.

Table 1

Values of the input parameters used in the case study.

Parameters	Description	Values
V_t	average train travel speed between stations	80 km/h
V_a	passenger walking speed	4 km/h
H_w	train headway at peak/off peak time period	10/15min
A_0	unit cost of access time	£14/hr
W_0	unit cost of waiting time	£14/hr
T_0	unit cost of in train travel time	£7/hr
S_0	unit cost of time spent searching for parking space	£14/hr
M_{t0}	unit operation and maintenance cost for train	£0.21/passenger-Km
M_{c0}	unit operation and maintenance cost for car	£0.26/passenger-Km
M_{b0}	unit operation and maintenance cost for bus	£0.16/passenger-Km
S_b	costs of building and equipping the stations with necessary facilities	£10 ⁶ /station
C_{T_s}	unit cost of building tunnel structure	£12x10 ⁶ /km
C_{Tr}	unit cost of track	£650/m
NS_{min}	minimum number of stations along each rail line	4 stations
NS_{max}	maximum number of stations along each rail line	7 stations
ΔS_{min}	minimum spacing between stations	800 m
ΔS_{max}	maximum spacing between stations	1500 m
Cn_{min}	minimum number of common/transfer stations along each rail line	1
On_{max}	maximum overlap rate between the rail lines	30%
ϕ_d	coefficient to account for demand pattern variations over weekdays and weekends	1
ϕ_p	coefficient of passenger cost	1
ϕ_o	coefficient of operator cost	1
ϕ_c	coefficient community cost	1
P_n	GA parameter-Population size	500
P_c	GA parameter-Crossover probability	0.7
P_m	GA parameter-Mutation probability	0.3
Gen	GA parameter-Number of generations	50

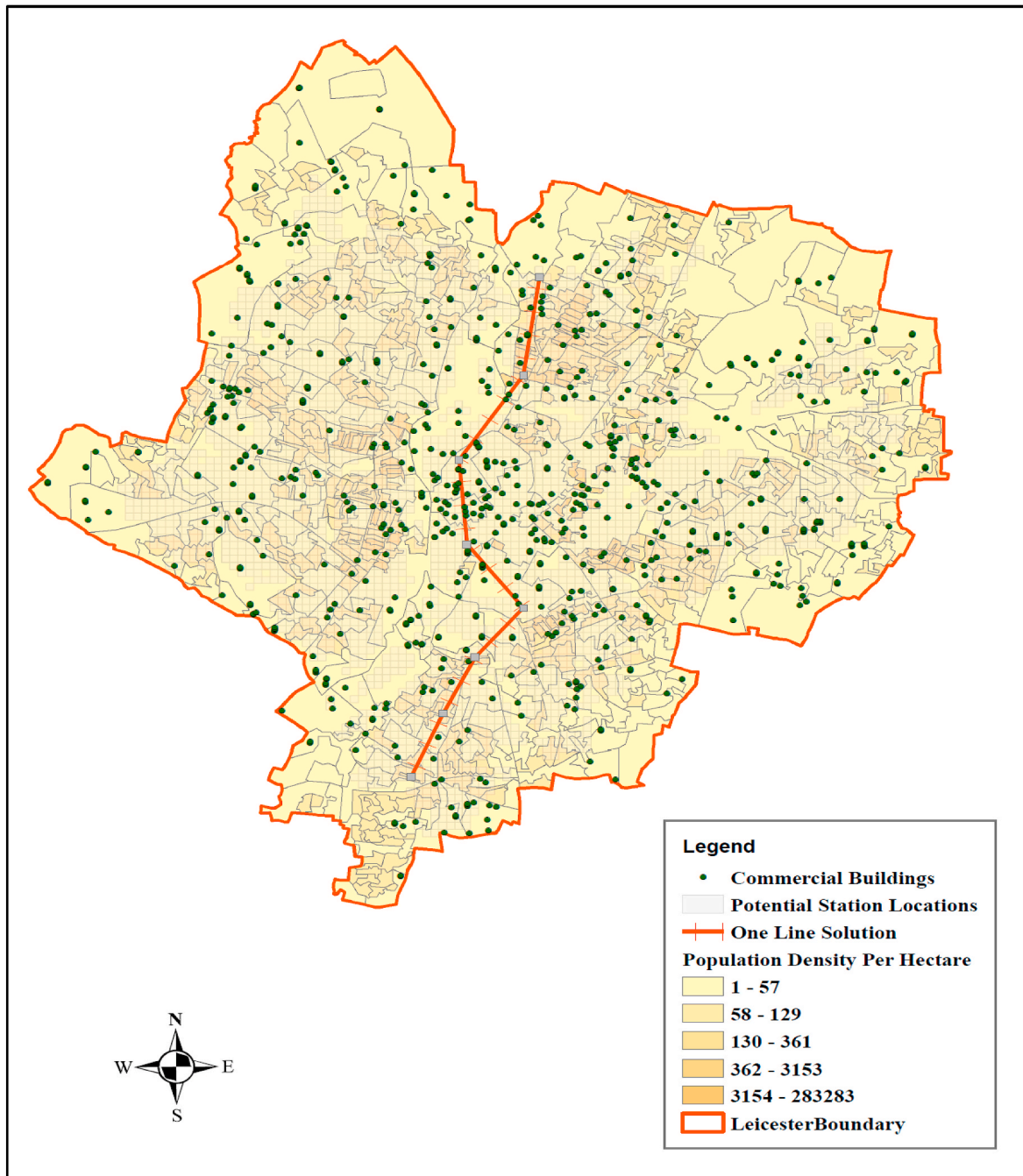
4. Case study

The effectiveness of the proposed model has been assessed using the City of Leicester (United Kingdom) as the study area. The required datasets were obtained from several agencies and incorporated into the GIS background database. These datasets include land use patterns, existing transport network, census, land values, future economic action plan, topography and origin-destination trip matrices. Two assumptions were made when applying the model: (1) the proposed rail system is underground/metro and; (2) stations are built at ground level while the line network is built in deep tunnels. The model outputs are presented in the following two sub-sections in line with the defined process in Fig. 2.

4.1. Generation of candidate stations

Following the procedure presented in section 3.1, and using the following input parameters, the model generates the candidate pool of potential station locations:

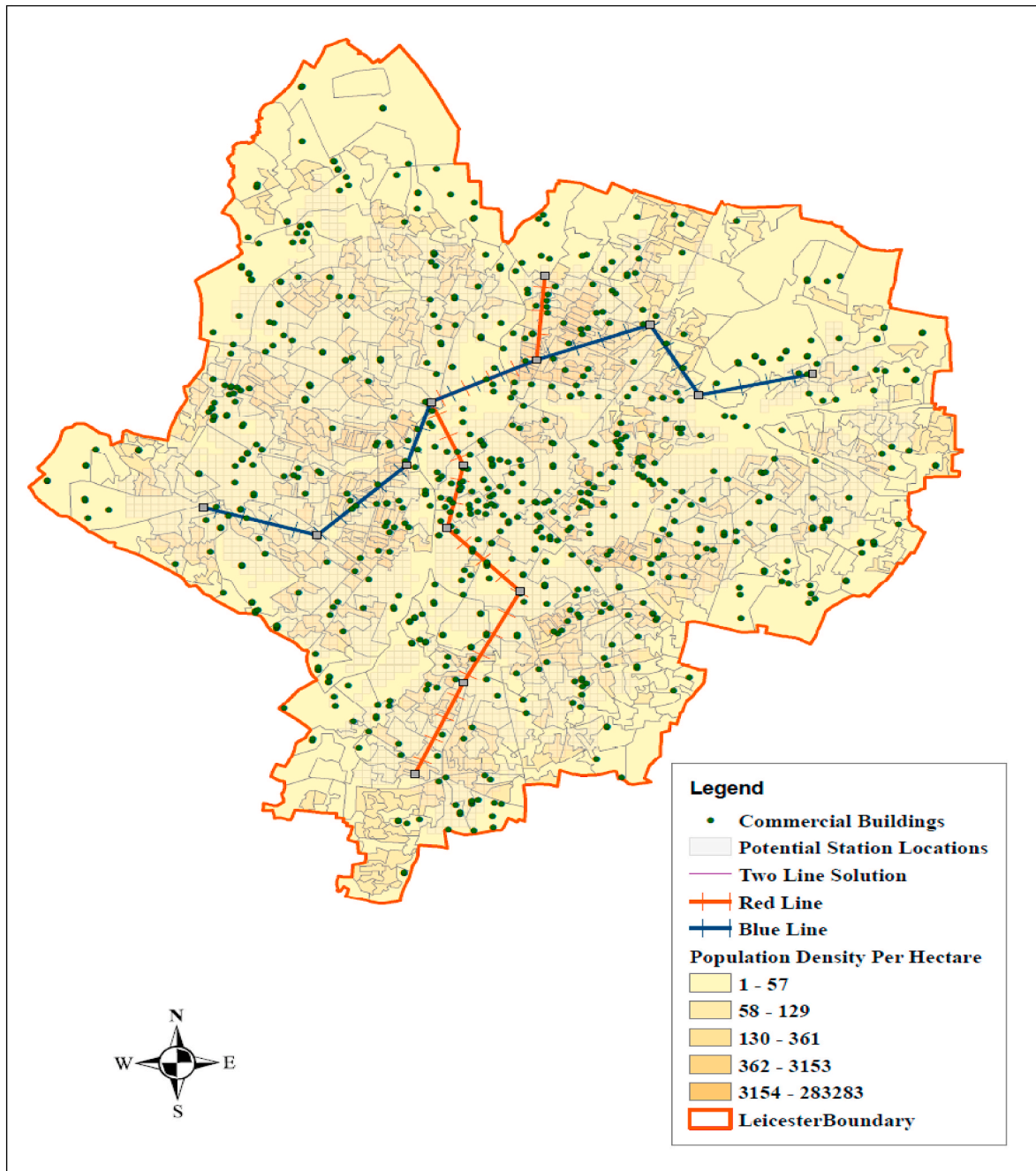
- 1 Station size:** the size of a typical, single-level track, underground/Metro station structure is 25×300 m for the respective width and length (Kaul, 2010). It is assumed, however, that the station concourse hall will be built at ground level to give the system a more attractive view. A station size (width and length) of 100×100 m was therefore adopted. These dimensions represent the station concourse hall width and length. Although it is larger than the required typical dimension for the station concourse areas, it is considered that accommodating some retail and coffee shops within this area would have many positive effects, such as increased system attractiveness and passenger convenience. In addition, more than one line may pass through some of the candidate stations after the line network is generated. This obviously requires a larger area for the station structure to accommodate additional services and associated facilities. Examples of such stations include Victoria and Kings Cross rail stations in London. Based on this station size, the City has a total of 7027 cells which represent potential rail station locations. For each cell, the defined requirement criteria were measured within a 500 m walking distance radius from the stations (O'Sullivan and Morrall, 1996; Bertolini, 1999; Vuchic, 2005; Evans et al., 2007; Zemp et al., 2011; Kamruzzaman et al., 2014).
- 2 Threshold values:** as discussed in section 3.1, the model allows rail transit system planners to define the threshold values of the requirement criteria based on the demographic, topological, economic and land use pattern characteristics of the served areas. For this case study, the 50th percentile of each of the calculated requirement criteria values was defined as its respective threshold value.



(a)

Fig. 7. Optimised station locations and associated line network: (a) One rail line solution; (b) two rail line solution; (c) three rail line solution.

After defining the proposed conditions for screening the study area and generating the candidate pool of feasible station locations, 4774 feasible station locations were found, as shown in Fig. 6. The results reveal that the candidate stations mainly cover the most densely populated and potential economic growth zones of the city. In addition, very high land values and environmentally sensitive areas were excluded from the search space. For simplicity, some features, such as sites of economic growth, bus stops, land values and restricted zones are not presented in the figure. Note that the validity of the grid based approach is ensured through a process that finds all station grids that intersect with environmentally sensitive areas e.g. historic buildings, national parks, woodlands, forests, rivers



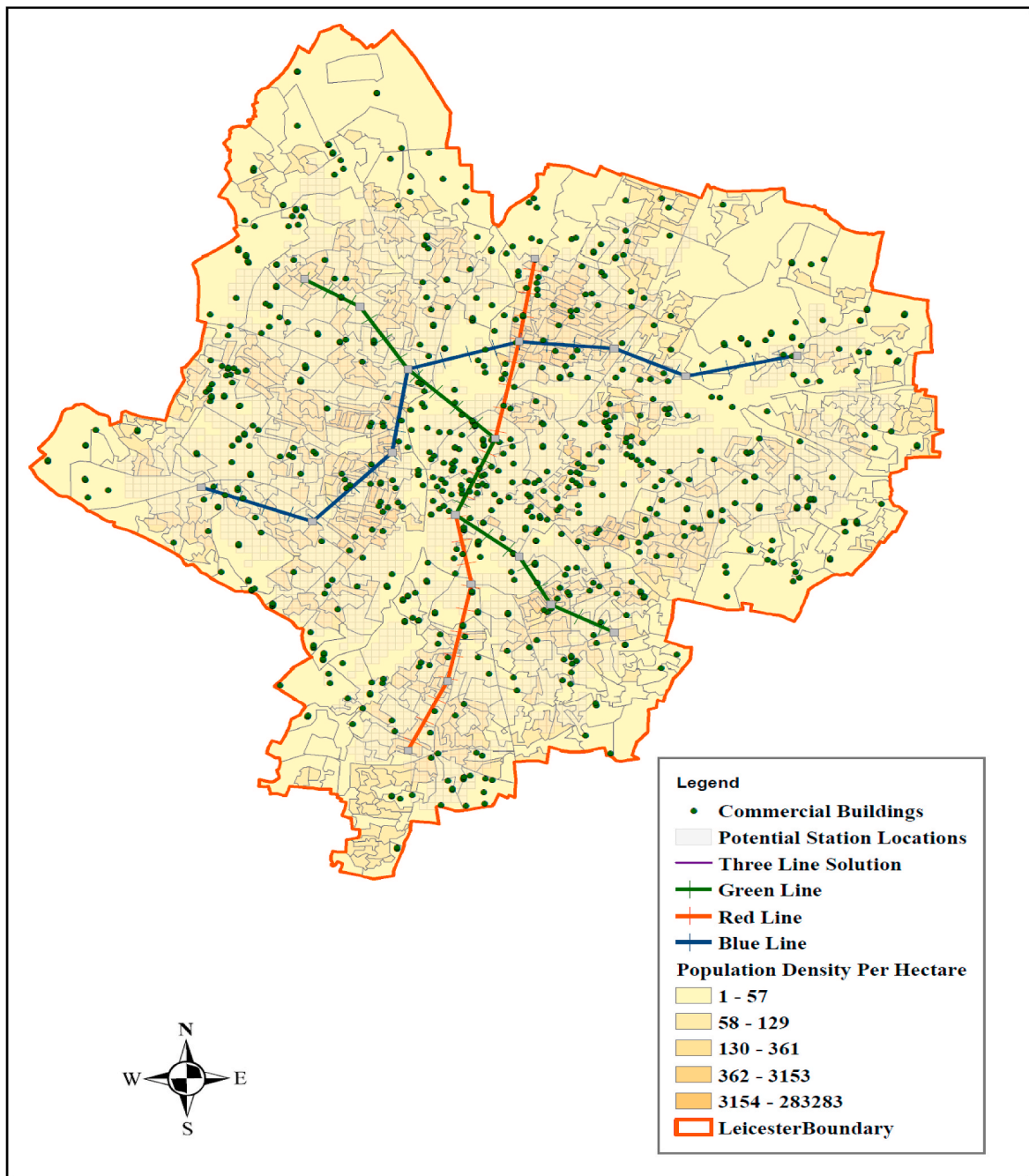
(b)

Fig. 7. (continued).

and sites of scientific interest) and excludes them from the search space of potential station location. In addition, users can impose a very high value for land acquisition and property damage leading to such locations being excluded during the optimisation process.

4.2. Optimisation of the station locations and associated line network

Once the set of feasible station locations is generated, the model proceeds to select the best subset from the identified feasible stations while simultaneously generating the best line network to link the selected stations using the parameters presented in Table 1. These parameters were derived from a combination of sensitivity analyses (see section 5), previous literature and transport analysis



(c)

Fig. 7. (continued).

guidance issued by the UK's Department for Transport.

It is assumed that the desired number of rail lines and terminal stations for each line are determined by rail transit planners based on the major traffic flow pattern of the area/city to be served. Accordingly, the proposed model generates the corridors within which the line network and associated stations are to be located based on the terminal stations. The basic topological configuration of the system is therefore bounded by these corridors. The range of topological configurations that can be considered is not limited in the sense that the proposed optimisation model can accommodate any basic topology and find the solution that best fits the desired objectives.

To examine the ability of the model to handle the different topological configurations of a rail transit system, three different

Table 2

Comparison of different threshold values of the requirement criteria on the optimised cost.

Threshold Values of the Requirement Criteria	Passenger Cost (M£)	Operator Cost (M£)	Community Cost (M£)	Total Cost (M£)	Daily Passengers
30th Percentile	−262.933	−91.516	233.086	−121.363	11602
40th Percentile	−309.767	−77.4735	256.355	−130.885	16545
50th Percentile	−304.305	−84.132	253.368	−135.069	16715
60th Percentile	−305.984	−104.567	255.731	−154.820	17685
70th Percentile	n/a	n/a	n/a	n/a	n/a

scenarios were applied to consider a proposed rail system consisting of a single rail line, two lines and three lines respectively. The terminal stations of every line in the three scenarios were determined based on the origin-destination trip matrices representing traffic flow patterns obtained from the Leicester City Council. Fig. 7a to c shows the results of these three test scenarios. In the first scenario, the proposed rail line links the south and north of the city. In the second scenario, one line links the south and north of the city while the other links the south-west and north-east areas. The three rail lines in the third scenario link the south and north, south-west and north-east, and north-west and south-east area of the city. Throughout the rest of this paper these lines are referred to as the Red Line (south/north); Blue Line (south-west/north-east) and Green Line (south-east/north-west) (see Fig. 7).

The results shown in Fig. 7 reveal that the model can efficiently generate different topological configurations. Fig. 7b, exhibits a star configuration while the proposed solution in Fig. 7c exhibits a combination of star and triangular configurations. It is also notable that the line alignments and associated station locations change as the overall size of the system changes, in order to minimise the total system cost while satisfying the required constraints. For example, compared to the first scenario, the alignment and station locations of the Red Line are different from both the second and third scenarios. Specifically, comparing the second scenario with the first, four stations starting from the fourth station in the Red Line (northbound from the southern terminus) have shifted towards the western side of the city centre, intersecting the Blue Line to the north-west in the second scenario. Accordingly, the overall alignment of the Red Line was adjusted towards these shifted stations as illustrated in Fig. 7b. Similarly, the Red and Blue Line alignments and associated station locations have changed in the third scenario compared to the second scenario. These results indicate that the proposed model can deal effectively with different system sizes when optimising station locations and associated line alignments.

5. Sensitivity analysis

This section first evaluates the importance of performing a feasibility analysis for potential station locations prior to the optimisation process to facilitate an efficient search process towards the optimal solution. It also investigates the effect of different threshold values for the requirement criteria on the optimisation search behaviour. Secondly, it examines the different parameters of the proposed genetic algorithm to determine the setting that best fits the framework of the rail transit system planning problem. Finally, it examines how simultaneous optimisation of all the rail lines reduces the total system cost compared to the optimisation of each line individually.

5.1. Impact of the determination of feasible station locations prior to the optimisation process

This section investigates how different threshold values for the requirement criteria affect the optimisation search space, with the aim of determining values that would not result in the exclusion of potential optimal station locations. Subsequently, this analysis evaluates how screening the study area for potential station locations prior to the optimisation process affects the performance of the proposed model in finding good solutions. The impact of the threshold values was evaluated by using five different threshold values for the requirement criteria, specifically 30th, 40th, 50th 60th and 70th percentiles, to identify potential station locations, and then to compare the optimised total system cost obtained for each of these threshold values. The reason for not including the 10th and 20th percentiles in this section is that there was little difference between these and the 30th percentile in terms of the number of feasible station locations. Table 2 presents the optimised cost for different threshold values of the requirement criteria for the two rail line system proposed in section 4.2.

The results show that the proposed optimisation model can find a better solution with higher threshold values, up to a point. The total cost of the system decreases by 27.56% from −121.363 M£ to −154.820 M£ when the threshold values of the requirement criteria are increased from the 30th to the 60th percentile. Compared to these two extremes, however, the reduction in the total cost of the system between the 40th and 50th percentiles is low, decreasing by just 3.19% from −130.885 M£ to −135.069 M£. This is because of the relatively low difference in the number of potential station locations. The potential station locations were 5163 and 4774 for the 40th and 50th percentiles respectively, and 5724 and 3856 for the 30th and 60th percentiles respectively. Note that the proposed model could not find a solution for the system when the threshold values were set at the 70th percentile. This is because, at this threshold value, there were insufficient potential stations to allow feasible alternative solutions to be generated (due to excessive narrowing of the search space). In this case, the number of potential station locations was 1749.

The results above indicate that screening the study area for potential station locations prior to the optimisation process plays an important role in directing the search towards the promising regions of the search space. When setting the threshold values of the requirement criteria, therefore, it should be ensured that optimal potential stations are not excluded. Based on the numerical results presented in Table 2 it can be concluded that setting the threshold values of the requirement criteria at the 60th percentile is efficient

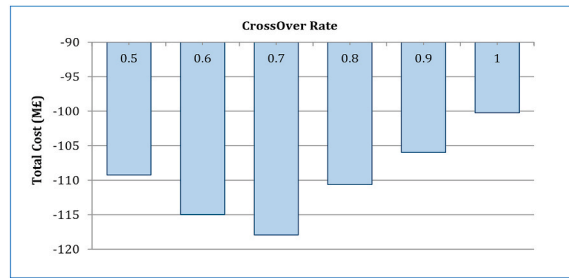


Fig. 8. Optimised total system cost with different crossover rates.

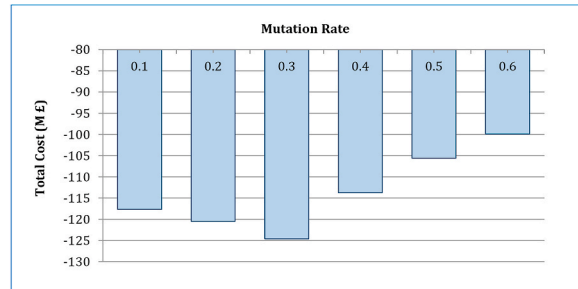


Fig. 9. Optimised total system cost with different mutation rates.

for finding a reliable and cost-effective solution for a two rail line system. Also, the results indicate that setting the threshold of the requirement criteria at low values or zero (i.e. no initial screening prior to the optimisation process) leads to the least cost effective solution.

The negative passenger cost values highlight the savings due to using the proposed rail system instead of using car or bus (see equation (10)). As for the operator cost, the negative sign refers to the total cost of operations and maintenance decreasing due to using the proposed rail system instead of using car or bus (see equation (18)).

5.2. Crossover rate sensitivity analysis

The sensitivity of the optimisation algorithm to the crossover rates (P_c) was assessed by evaluating the optimal total system cost for six crossover rates in the typical range of 0.5–1. The results in Fig. 8 show that the algorithm can find a better solution with a higher crossover rate, but only to a certain extent. The total cost of the system decreased by 8.72 M£ when the crossover rate increased from 0.5 to 0.7. Further increments in crossover rate beyond 0.7, however, tend to be inefficient in improving the solutions. The results also show that the best solution was obtained with a crossover rate of 0.7. As discussed in Section 3.2.3, this can be explained by the fact that a high P_c value creates more new offspring/children and subsequently, leading the algorithm to explore more of the search space and mitigates the risk of trapping in local optima. However, a very high P_c value may destroy good chromosomes (i.e. alternative solutions), leading to suboptimal solutions.

5.3. Mutation rate sensitivity analysis

The sensitivity of the optimisation algorithm to different mutation rates (P_m) was assessed by evaluating the optimal total system cost for five mutation rates in the 0.1–0.6 range. The results in Fig. 9 show that the algorithm can find better solutions with a higher mutation rate, but only to a certain extent. As depicted in Fig. 9, the total cost decreases by about 7 M£ when the mutation rate is increased from 0.1 to 0.3. The optimisation algorithm tends to degrade the quality of the solution with further increments in the mutation rate beyond 0.3, however. This is because a very high mutation rate tends to lead to a significant diversification of the search space which may result in pulling good solutions away and thus preventing the algorithm from converging at any optimal solution. The results show that the optimum solution was obtained with a mutation rate of 0.3.

5.4. Comparison of individual and simultaneous optimisation

Individual optimisation methods optimise the alignment and associated station locations of each line separately. On the other hand, simultaneous optimisation methods look at the holistic optimisation of the rail network by optimising the alignment and station locations of all the lines simultaneously. This section compares individual and simultaneous optimisation methods for planning a rail transit system. Table 3 presents a comparison of the optimisation results of these two methods on a rail transit system with two rail

Table 3

Comparison of independent and simultaneous optimisation for a two rail line system.

Two Lines (Red-Blue)Lines	Simultaneous Optimisation	Individual Optimisation			Difference %
		Red Line	Blue Line	Total	
Passenger Cost (M£)	−304.305	−144.750	−116.975	−261.725	−16.27
Operator Cost (M£)	−84.132	−33.155	−41.290	−74.445	−13.01
Community Cost (M£)	253.368	131.489	125.211	256.700	1.30
Total Cost (M£)	−135.069	−46.416	−33.054	−79.470	−69.96
Line Network Length (km)	18.131	9.123	7.988	17.111	−5.96
Total number of Stations	14.000	7.000	8.000	15.000	6.67
Daily Passenger	16715	5289	3874	9163	−82.42

Table 4

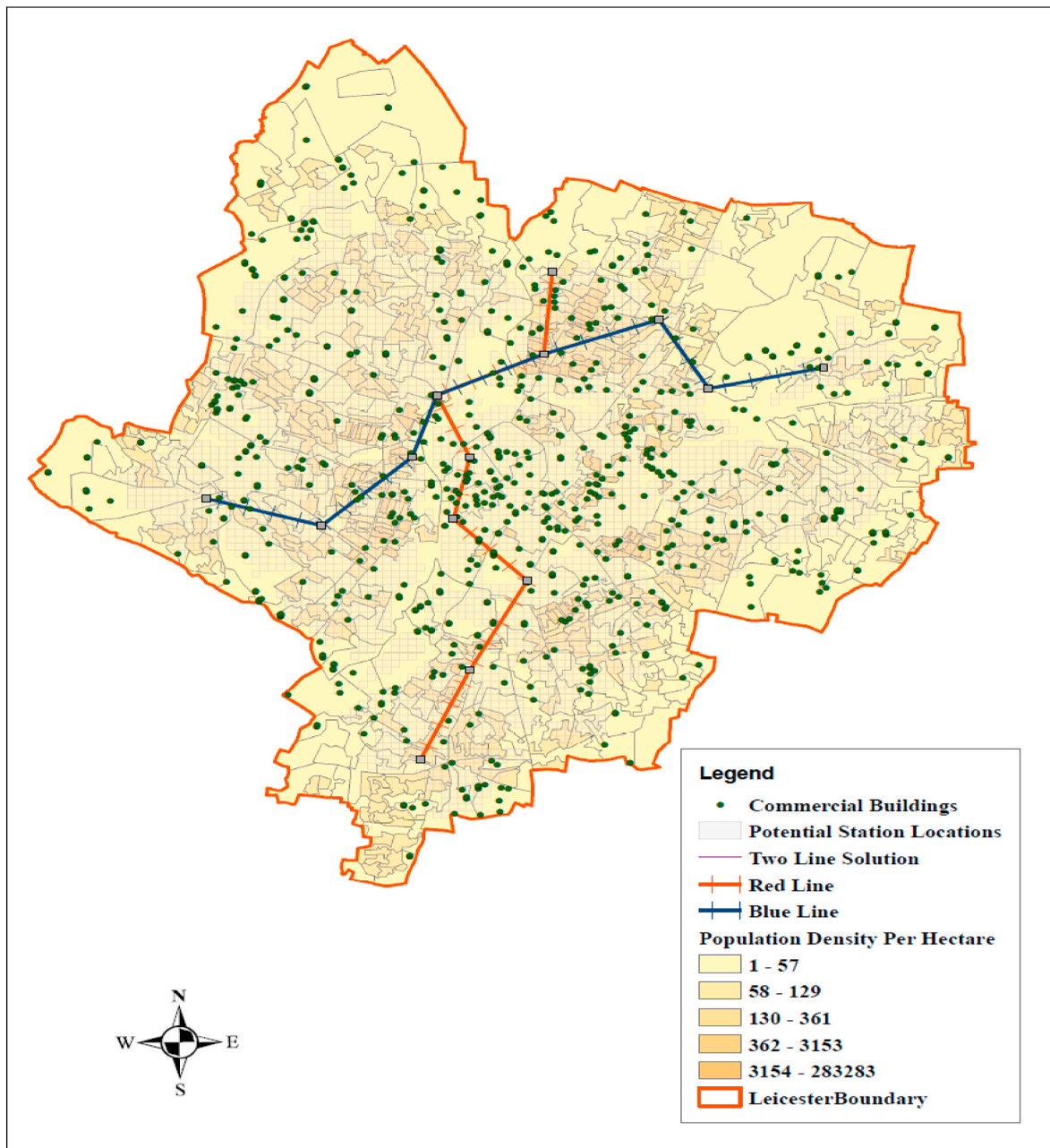
Total cost of the solutions found manually and the solutions found by the proposed model.

Solutions Found Manually							
Solutions No.	Total cost (M£)	Solutions No.	Total cost (M£)	Solutions No.	Total cost (M£)	Solutions No.	Total cost (M£)
1	−8.573	26	−4.972	51	−9.520	76	−3.618
2	−9.897	27	−9.416	52	−13.328	77	−8.464
3	−5.800	28	−20.670	53	−9.711	78	−25.742
4	−11.534	29	−8.259	54	−5.707	79	−13.503
5	−8.376	30	−11.444	55	−7.733	80	−8.998
6	−7.522	31	−11.242	56	−7.099	81	−19.268
7	−10.647	32	−8.763	57	−6.069	82	−8.306
8	−6.775	33	−8.152	58	−9.098	83	−18.730
9	−10.333	34	−10.442	59	−9.999	84	−21.841
10	−11.840	35	−13.031	60	−9.245	85	−29.186
11	−8.462	36	−7.874	61	−12.909	86	−8.262
12	−5.381	37	−8.625	62	−4.330	87	−5.577
13	−13.788	38	−10.831	63	−11.721	88	−11.791
14	−12.375	39	−5.239	64	−7.029	89	−12.291
15	−13.570	40	−7.160	65	−15.581	90	−17.801
16	−11.661	41	−9.485	66	−12.418	91	−10.476
17	−13.340	42	−9.029	67	−9.210	92	−7.666
18	−7.676	43	−6.055	68	−9.917	93	−6.404
19	−9.462	44	−7.217	69	−8.313	94	−9.054
20	−7.633	45	−12.490	70	−10.942	95	−9.724
21	−9.222	46	−7.230	71	−8.622	96	−11.050
22	−12.352	47	−14.689	72	−17.188	97	−8.668
23	−11.740	48	−7.176	73	−13.831	98	−4.744
24	−12.043	49	−19.969	74	−7.258	99	−11.359
25	−5.028	50	−7.439	75	−8.860	100	−9.035
Solutions Found by the Proposed Integrated Optimisation Model							
Solutions No.	Total cost (M£)						
1	−36.306						

lines.

Compared to the individual optimisation, simultaneous optimisation significantly reduced the total cost, by about 70% from −79.470 M£ to −135.069 M£, with 82.42% more daily passenger numbers. Also, the simultaneous optimisation decreased the community cost from 256.7 M£ to 253.368 M£, passenger cost from −261.725M£ to −304.305 M£ and operator cost from −74.445 M£ to −84.132 M£. These numerical results prove the advantage of simultaneous over individual optimisation.

Fig. 10a presents the solution generated from the simultaneous optimisation, while Fig. 10b presents the solution generated from the individual optimisation. In general, although both solutions exhibit a similar star-shaped topological configuration, the length of the total line network, and the station numbers, as well as station locations and line alignments generated from the two optimisation methods, are different. The total line network length and station numbers generated from the simultaneous optimisation were 18.131 km and 15 stations, while with the individual optimisation they were 17.111 km and 16 stations. It is important to note that the locations of the first intermediate stations from the terminal stations did not change for either of the optimisation methods but that the locations of the other intermediate stations and their associated line alignment changed. The stations in the individual optimisation and the associated line alignment were generated to attract the maximum possible number of rail passengers for each line individually without considering the availability of other lines in the system and possible passenger transfer between these lines. In contrast, the simultaneous optimisation method selected the station locations to attract the maximum possible number of railway passengers for all the lines in the system, taken together and considering the connectivity between the lines. As depicted in Fig. 10a, the generated rail lines intersect at two different stations in the north-west and northern sides of the city centre. This, in turn, resulted in a significant



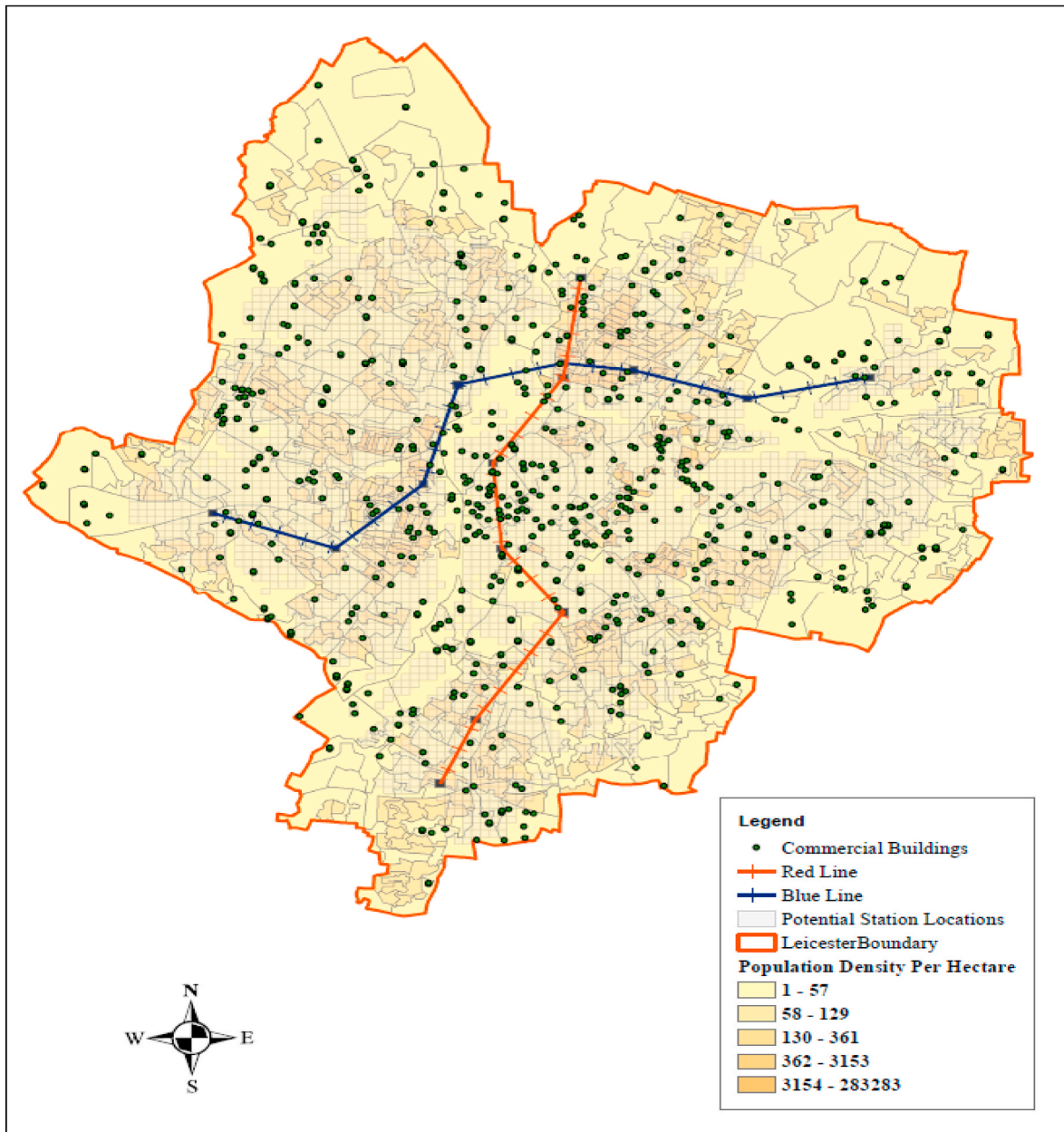
(a)

Fig. 10. Optimisation methods of a two rail line system: (a) Simultaneous optimisation method; (b) individual optimisation of a two rail line system.

increase in passengers attracted to use the system, and subsequently decreased the total cost significantly compared to the solution generated from the individual optimisation method, where the lines, although crossing, do not intersect at a station. Analysing the benefit of the integrated approach to a transfer function-based assembly of the individual optimisations is not considered in this paper but it will be included in future work.

6. Goodness evaluation for the best solution

This section assesses the quality of the solution from the proposed integrated optimisation model. Selecting a set of station locations



(b)

Fig. 10. (continued).

and line alignments linking the stations between two terminal stations can generate a large number of possible solutions and local optima, which makes it impractical to ascertain the best solution. Thus it is very difficult to prove the goodness of the solution found by the proposed model. A review of the literature highlighted that none of the previous studies in this area have been reliably able to identify an optimal solution to test the GA solution against for the full network. This section attempts to address this problem of testing through designing a novel experiment to test the goodness of the model based on a single line. The experiment is initiated by finding a solution to the problem using the proposed model and then generating a set of possible solutions to the problem manually that are believed to be close to the optimal solution. Thereafter, the fitness values of the solution found by the proposed models are compared with those generated manually. The experiment considers the solution found by the model to be good if the fitness value of the solution found by the model is higher than that of all other possible solutions found manually.

Various iterations of this experiment were conducted in order to test the goodness of the model for most of the terminal pairs

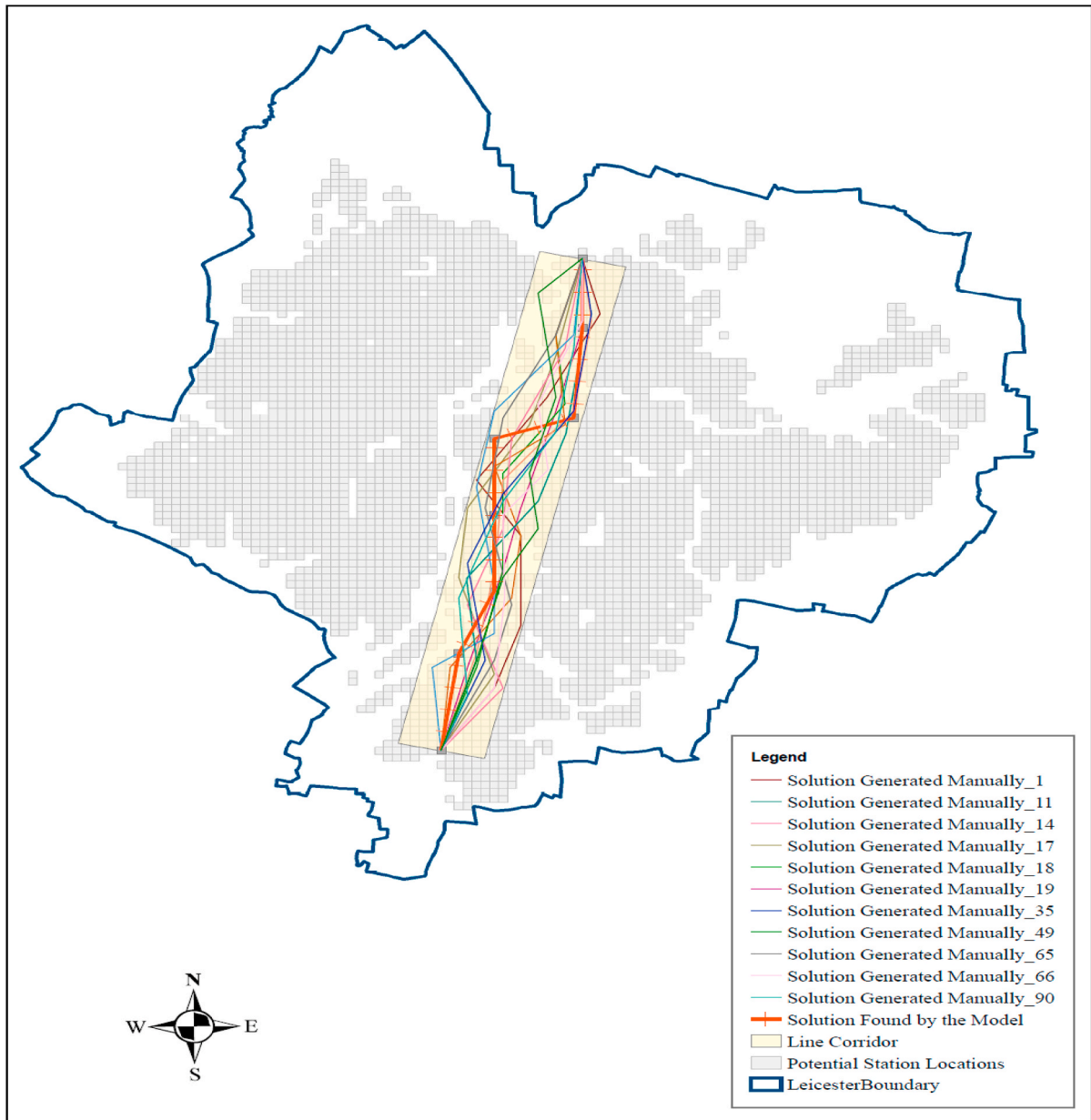


Fig. 11. Illustration of the Proposed Model Validation.

identified in section 4.2. Fig. 11 illustrates the solution found by the model, and some of the manually generated solutions, for the south-north terminal station pair. The whole set of solutions are not presented for clarity and ease of comprehension.

Note that the width of the corridor within which the station locations and line alignment are to be determined was set at a relatively small value (1 km), as depicted in Fig. 11 in order to make the test more practical. Increasing the width of the corridor would improve the test robustness but will increase the required manual effort. Table 4 presents the fitness in terms of the total cost of the solution found by the proposed model and the other possible solutions found manually.

As can be seen in Table 4, the solution found by the proposed model possesses the highest fitness value compared to the other possible solutions.

7. Conclusions

This paper has developed an integrated optimisation model for planning a rail transit system accounting for the various local conditions and multiple requirements that arise from the different stakeholders. The model considers the complex correlations and

interactions between rail transit station locations and line alignments by integrating these two intertwined components within a single optimisation process, a task which has been largely neglected in the existing literature, and which also enables simultaneous optimisation of multiple rail lines. This reduces the total system cost significantly, and thus achieves a cost-effective solution. Furthermore, the developed model formulations address the essential trade-offs between maximum rail system usage and minimum passenger travel time, on the one hand, and the minimum construction cost of the system on the other hand, while complying with various constraints.

The City of Leicester (UK) has been used as a case study to examine the effectiveness of the model in finding a solution that best fits the desired objectives. An extensive numerical study was also conducted to examine the importance of the initial identification of the potential station locations in directing the search process towards optimal solutions; and to compare individual and simultaneous optimisation methods for planning a rail transit system. The results show that performing a feasibility analysis for potential station locations prior to the optimisation process can improve the performance of the model by directing the search to explore the more promising regions in the search space. Furthermore, the numerical results confirm the advantage of the simultaneous optimisation of multiple rail lines compared to individual line optimisation. Simultaneous optimisation reduced the total cost by about 70% over individual line optimisation. Future work will involve further sensitivity analysis associated with the limitations of some of the assumptions made such as the trip distribution pattern between the traffic analysis zones of the study area not changing after building a proposed rail system. Future work will also analyse the benefit of the integrated approach to a transfer function-based assembly of the individual optimisations.

Declaration of competing interest

None.

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